

Sopra Steria Consultancy Report: Television Scheduling and Advertising

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November 2024

Group Project for Methodology, Modelling and Consulting Skills

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Executive Summary

Channel A engaged our team to develop a comprehensive solution for optimizing their television programming and advertising strategy in order to maximize viewership. As the audience base of the traditional television market has plateaued and is expected to stagnate for the foreseeable future, leveraging mathematical optimization models to maximize the value extracted from existing viewers has become pertinent. Our approach revolved around splitting the problem into two key components:

1. First, we developed a computationally efficient movie scheduling model that determines an optimal schedule to maximize total viewership for the full 12 weeks with a typical runtime of a few minutes per week. The scheduling model avoids movies shown by competitors (both within the current week and in previous weeks) and at the same time, chooses a unique schedule every week to avoid repeating the same schedule. After analysing the output of the scheduling model, we were able to extract key insights for content prioritization for channel A which can be summarised in the following three key takeaways:
    - Diversify genres shown throughout the day to cater to different preferences
    - Focus on newer, more expensive productions as they perform better
    - Prioritize content for adults and tactically cater to children and retirees given their size
2. The second component focused on advertising strategy, which is the primary revenue driver for commercial television channels. We devised a market-driven framework that is also computationally efficient (running in a matter of seconds for every week) for evaluating both the cost-effectiveness of acquiring advertisement slots from competitors and the impact of using our own advertising slots. Our analysis identified three major findings that Channel A can capitalize on:
    - Purchase competitors' advertisement slots that are under-priced relative to expected viewership
    - Incorporate viewership into the pricing of our own advertisement slots
    - Target advertisements based on expected genre similarity to maximize conversion

Implementing only our scheduling model allows Channel A to maintain its market share despite fierce competition and stagnant television market. Combined with our market-driven advertising strategy, our model achieved a minimum profit margin of 60% every week over the entire 12 week period and we are confident that it will allow Channel A to dominate the United Kindom's television market.

Contents

1 Introduction **1**

2 Methodology **1**
2.1 Movie scheduling . . . 2
2.2 Computationally feasible approximation . . . 3
2.3 Advertising strategy . . . 4
2.3.1 Why does advertising matter? . . . 4
2.3.2 What is the value of an ad slot? . . . 4
2.3.3 How much viewership do adverts convert? . . . 4
2.3.4 How do we calculate $c_{k,m}$? . . . 5
2.3.5 Which adverts should we buy? . . . 5
2.3.6 How to establish the value V of a unit of ad slot viewership . . . 6
2.3.7 Potential objections to the calculation of the value of a viewer . . . 7

3 Results **7**
3.1 Scheduling results . . . 7
3.1.1 Viewership change . . . 7
3.1.2 Market share evolution . . . 7
3.2 Content strategy . . . 8
3.2.1 Genre and demographic focus . . . 8
3.2.2 Release dates and licensing costs . . . 9
3.3 Advertising and pricing strategy . . . 9

4 Conclusion **10**

Appendices **12**

A Advertisement planning **12**

B Source code **12**

C Full advertisement strategy results **13**

1 Introduction

Television scheduling is a well-researched area with early beginnings that include combinatorial optimization to maximize ratings [Hor80] and mixed-integer, generalized network-based flow models to devise an optimal schedule [RAS98]. The importance of such research is pertinent especially considering the current market dynamics of the television industry. The traditional television market remains one of the most stable and resilient markets. Recent reports predict that the traditional television market will remain constant for the foreseeable future [Sta24]. Despite the rapid rise of online streaming platforms, the core television audience has demonstrated remarkable consistency. The stability presents a clear opportunity to capitalize on the large, stable viewership base, since there is limited potential for expanding the audience. By leveraging mathematical optimization and applying operational research techniques, we can optimize scheduling to extract maximum value for the shareholders.

As consultants for channel A, our task is to optimise their content scheduling and advertising strategy. In content scheduling, the goal is to choose the best permutation of movies from a database of 5,920 movies by considering factors such as genre, release date, and scaled popularity for different demographics. There are 3 other competitor channels: *0*, *1*, and *2*. Each channel streams movies continuously from 7:00am to 12:00am with one advertisement break approximately every 30 minutes. The challenge lies in choosing the optimal mix of content while factoring in competitors' schedules and the dynamics of audience viewership through time.

The advertising strategy component adds significant complexity. It is simultaneously at the heart of our business model as *we assume all revenue is generated through ad sales.* Some television channels do not aim to make money through advertising (including publicly funded broadcasters such as the BBC) but their purpose and funding models are significantly different from our own. Subscriptions to packages of TV channels (e.g. Sky TV) also represent a very different product than our own. Practically, given the availability of such services and Netflix-like on-demand video, it is implausible that a customer would pay a subscription for a single channel offering a diet of movies over which they have no control. From a theoretical standpoint, the modelling of subscription services brings complexities which lie out of the scope of our approach. These reasons justify the approach we have taken.

A well-crafted advertising strategy is critical to generate value to the shareholders. Advertising on competitors' channels involves costs but has the potential to convert viewers into our channel for the movie being advertised, thereby increasing our future viewership. Promoting a horror movie during a competitor's showing of a horror movie is likely effective, whereas placing it during a romantic comedy might not yield results. On the other hand, we can advertise on our own channel which is free but limits reach to the existing audience and has an opportunity cost of the potential revenue we could have made selling that advertisement slot.

In summary, the following outcomes were achieved and form the content of this report:

- We developed a computationally feasible approximation to a comprehensive mathematical model to generate optimal schedules efficiently. Even with this standalone scheduling model (without considering advertisements), we successfully maintained our market share against competitors.
- In the second part, we devised an approach to the purchase and sale of ad slots on our own and competitor channels which drives an increase in viewership on our own channel (at low cost) and corresponding increase in ad revenue and profitability.

After detailing the approach taken to both these parts, the results of our scheduling and advertising approaches are presented and analyzed. Our key recommendations are also summarised.

2 Methodology

Our approach to the problem is divided into two parts: optimising our movie schedule and determining an advertising strategy to maximize viewership and profit (defined as advertising revenue minus licensing fees).

In the first part, we allocate movies to time slots to maximize viewership across all demographics while ensuring unique movies (compared to our competitors) and through time. We do this by developing a sophisticated mathematical formulation that ensures only one movie can occupy any given time slot, the total time allocated to each movie is *exactly* equal to its duration with ads, and properly schedule a movie in consecutive slots.

In the second part, we establish a fair market price for adverts (based on their viewership), model the rate at which adverts convert viewers into our movies, identify which ad slots and which movies are potentially profitable (based on their ability to generate additional viewership we can convert into revenue), and use a carefully constructed algorithm to ensure that the interaction effects of choosing a selection of adverts are accounted for. We thus identify which adverts on competitor channels to purchase and which on our own channel to sell. The selection of such adverts constitutes a strategy which we determine will return a profit.

By integrating these two components—an optimized content schedule and a profit-driven advertising strategy—the strategy indicated by our model indicates a route to realistic, profitable performance.

2.1 Movie scheduling

In this section, we first propose a model to determine the optimal movie schedule for our channel. We aim to develop a optimal one-week movie schedule that incorporates ad-time-inclusive running durations. Given the complexity of the decision-making problem involving advertisement slots, as described in Section A, we employ a separate strategy to optimize decisions regarding internal and external advertisements—specifically, whether to sell internal ad slots or purchase external ad slots.

The parameters used in the model are as follows:

- $\mathcal{T}$: the set of time slots in one day.
- $\mathcal{M}$: the set of movies.
- $\mathcal{D} = \{1, \ldots, 7\}$: the set of days.
- δ_m: the number of time slots occupied by movie m.
- $M_m = |\mathcal{T}| - \delta_m + 1$: the latest possible time slot in which movie m can start.
- $f_{m,t,d} \in \mathbf{R}$: the estimated expected viewership for displaying movie m during time slot t on day d, calculated as the sum of the baseline viewership of three audience groups (children, adults, and retirees), multiplied by the movie's scaled up popularity and prime time factor.

The decision variables are:

- $x_{m,t,d} = 1$: if movie m is shown at time t on day d; 0 otherwise.
- $y_{m,d} = 1$: if movie m is shown on day d; 0 otherwise.
- $s_{m,d} \in \mathbf{N}$: start time slot of movie m on day d.
- $e_{m,d} \in \mathbf{N}$: end time slot of movie m on day d.

The model is:

$$\max \sum_{m\in\mathcal{M}} \sum_{t\in\mathcal{T}} \sum_{d\in\mathcal{D}} f_{m,t,d} x_{m,t,d} \tag{1a}$$

$$\text{s.t.} \sum_{m\in\mathcal{M}} x_{m,t,d} = 1 \qquad \forall t \in \mathcal{T}, d \in \mathcal{D} \tag{1b}$$

$$\sum_{t\in\mathcal{T}} x_{m,t,d} = y_{m,d}\delta_m \qquad \forall m \in \mathcal{M}, d \in \mathcal{D} \tag{1c}$$

$$e_{m,d} - s_{m,d} = y_{m,d}\delta_m \qquad \forall m \in \mathcal{M}, d \in \mathcal{D} \tag{1d}$$

$$e_{m,d} \geq (t+1)x_{m,t,d} \qquad \forall t \in T, m \in \mathcal{M}, d \in \mathcal{D} \tag{1e}$$

$$s_{m,d} \leq t x_{m,t,d} + (1 - x_{m,t,d})M_m \qquad \forall t \in T, m \in \mathcal{M}, d \in \mathcal{D} \tag{1f}$$

The objective (1a) is to maximize the total viewership over one week on the channel, with constraints:

- (1b) ensures that at each time slot t on any day d, exactly one movie is shown.
- (1c) ensures that the total time allocated to a movie m on day d equals its duration δ_m.

- (1d) enforces the difference between end time and start time corresponds to the movie's duration.
- (1e) specifies that if $x_{m,t,d} = 1$ the end time must be the latest $t + 1$ such that $x_{m,t,d} = 1$.
- (1f) ensures that if $x_{m,t,d} = 1$, $s_{m,d}$ reflects the earliest t for which $x_{m,t,d} = 1$, marking the actual beginning of the movie's runtime.

2.2 Computationally feasible approximation

While the mathematical model developed is robust and comprehensive, solving it within practical computational constraints—such as a one-hour runtime—proved infeasible due to the sheer number of decision variables and constraints involved. To address this, we adopted several strategies to balance computational efficiency with the relative optimality of the results.

Aggregating time intervals

We reduced the granularity of the scheduling horizon $\mathcal{T}$ by aggregating 5-minute time slots into 30-minute intervals. This approach reduced the problem size by a factor of six while preserving the model's core structure. This adjustment aligns with the practice of calculating movie runtime, including advertisements, in 30-minute blocks. Advertisements also follow the same structure, with one ad in every 30 minutes.

While this approximation introduces minor information loss by averaging viewership across six 5-minute slots, its impact is limited. Given that viewership is unlikely to exhibit erratic changes within such short intervals, deviations can reasonably be attributed to random noise. Consequently, the 30-minute aggregates serve as a reliable proxy for viewership trends over the scheduling horizon.

Compressing the movie database

The size of the movie database $\mathcal{M}$ was reduced by applying a filter based on (near) Pareto-efficiency/non-domination. Movies were pruned (i.e. removed) if there was more than a threshold number of movies with the same or lower license fee but with the same or higher popularity across all three demographic groups (adults, children, and retirees). This considerably reduced the set of movies, while retaining a variety of high-potential content for optimization.

While this approach could reduce the movie database by a factor of more than 200 with a threshold of 0, the threshold parameter was adjusted to produce reductions of closer to a factor of 60, to ensure sufficient movies were left over. Typical threshold values were between 0 and 2, resulting in a typical filtered database of between 100 and 200 movies each week.

Ensuring a diverse movie schedule every week

To ensure that our channel presents a unique value proposition to viewers, we incorporated heuristic-based pre-processing of the movie database. This pre-processing operates on the full movie database each week, prior to pruning.

Based on the observation that no movie was shown on more than one competitor channel within a single week, we exclude a movie from scheduling if appears within any competitor's schedule in the coming week.

We also excluded movies shown on *our* channel in any prior week. Our approach will therefore never repeat a film over the planning period.

Optimizer settings

The target MIP gap was 5% and the timeout was 600 seconds (i.e. 10 minutes). With one exception, the solver achieved the MIP gap well within the timeout for all weeks (typically taking only 1 or 2 minutes per week). The one exception was the week 2024-10-29 to 2024-11-05 in which the optimizer achieved a gap of only 6.4% after the timeout.

2.3 Advertising strategy

2.3.1 Why does advertising matter?

Advertising serves two purposes in our model:

1. **Boosting Viewership**: Internal advertising slots on our channel and external slots on competitors' channels can be utilized to promote movies from our upcoming schedule, increasing our channel's viewership.

2. **Generating Revenue**: Internal slots can also be sold to external entities, including other channels advertising their own content or commercial advertisers promoting products and services.

Under the assumption that advertising is the sole revenue source for our channel and its competitors, a coherent approach necessarily involves commercial adverts. Indeed, viewing the 'eco-system' consisting of our own channel and its competitors as a whole, there are cashflows out for license fees but in the absence of external ad revenue, there is no cashflow in: payments between channels are only internal to the eco-system.

Our approach to ad slot sales on our channel therefore permits both sales to competitor channels and sales to commercial partners. We do not, however, distinguish between them which leaves open for further consideration the effect of selling our own ad slots to competitors on our own viewership.

2.3.2 What is the value of an ad slot?

Since advertisements are our sole revenue stream, determining how to price them is essential.

In practice, many factors influence the price a commercial partner is willing to pay for an advertisement slot. Chief among these is the number of viewers the advertisement reaches within the relevant market segments.[Med24]. Commercial partners assess whether the advertising cost is allowable—that is, whether the cost-per-view is lower than the derived benefit.

We therefore define the market value of an ad slot to be $\sum_{g \in G} V_g \nu_g$, where V_g is the value of a viewer in demographic group g and ν_g is the expected number of viewers of the slot who fall into g. The demographic groups $g \in G$ partition the entire potential viewership into disjoint sets so this is a clean market segmentation. Once the values of V_g are established, the price of an ad slot can be immediately calculated. (In context, these values represent how much it is worth to an advertiser for their advert to be seen by a child, adult, or retiree, respectively.) .

We simplify this model in a way that still represents an improvement on the ad-pricing strategies of our competitors, by determining an average value V per viewer of an ad slot and approximate $\sum_{g \in G} V_g \nu_g$ by $V\nu$, where ν is the total viewership of the slot. This simplification is justified by consideration of the available data and is discussed later. We consequently price our own ad slots at the market price $V\nu$.

On our own channel, we choose to place ad slots only within single movies (rather than between movies) and judge that this offers the greatest clarity to potential partners about the demographic groups they can target through our ads.

2.3.3 How much viewership do adverts convert?

Placing an advert for a movie is expected to increase the viewership of that movie. The increase in viewership $\Delta\tilde{\nu}_m$ of movie m (conversion) due to the advertising it in ad slot k is determined by three factors: the viewership of the ad slot, the proportion of those viewers susceptible to conversion, and the effectiveness of the advert.

It is only possible to convert viewers to watch a movie if they have not already decided to watch it. Consequently we define the remaining convertible viewership as $\tilde{r}_m = 1 - \tilde{\nu}_m$, where $\tilde{\nu}_m$ is the viewership of movie m. For definiteness, and to ensure the expected viewership remains below 1 even after advertising, we set $\tilde{\nu}_m = \max\{\nu_t : \text{movie } i \text{ is showing in time slot } t\}$, so we consider the smallest convertible population. (In practice, expected viewership does not vary wildly within the timeframe of a single movie.)

We also define $c_{k,m}$ to be the effectiveness of our advert. It is the proportion of remaining convertible viewership that an advert converts: *independent of any other ad slot purchases*, buying only ad slot k to advertise movie m would increase the viewership of movie m by

$$\Delta\tilde{\nu}_m = c_{k,m} \cdot \tilde{r}_m \cdot \nu_k$$

across all the slots in which movie m is shown.

Crucially, when we advertise movie m (increasing its viewership) the value of $\tilde{r}_m$ will decrease. We identify in Section 2.3.5 how to sequentially select the ad slots and movies to advertise, taking this dependence into account.

Ideally, a more sophisticated version of this approach would disaggregate these conversions by demographic group. With further time and resource, this would be a promising avenue for further investigation.

2.3.4 How do we calculate $c_{k,m}$?

Building on the approach currently used by the Client, we determine the overlap of the genre(s) of the movie m we seek to advertise and of the movie(s) shown around ad slot k. From this overlap score, the Client determines a stochastic conversion rate, as a way to account for the uncertainty of how well-received an ad is likely to be. The expectation of this stochastic conversion rate (which is not in fact equal to the genre overlap score, and which we set to 0 when movie m plays before ad slot k) we denote by $E_{k,m}$. The calculation of the stochastic conversion rate using cosine similarity is known to the Client and the its expectation is derived from elementary statistics.

We then include scaling based on the *popularity* ρ_m of the movie m. This is an improvement because, in reality, popular movies will be more attractive to ad viewers than unpopular movies. In the determination of our advertising strategy, we therefore rely on the *expected* conversion rate for movie m in ad slot k of

$$\bar{c}_{k,m} = E_{k,m} \cdot \rho_m.$$

Consequently, the *expected* conversion is

$$\mathbb{E}[\Delta\tilde{\nu}_m] = \bar{c}_{k,m} \cdot \tilde{r}_m \cdot \nu_k = E_{k,m} \cdot \rho_m \cdot \tilde{r}_m \cdot \nu_k,$$

accounting for the three factors we identified as relevant: advert effectiveness $E_{k,m} \cdot \rho_m$, proportion of the population still convertible $\tilde{r}_m$, and the viewership of the ad slot ν_k.

2.3.5 Which adverts should we buy?

The simplest characterisation of ad slots which are profitable for us to buy is as *those which generate enough additional viewership to justify their cost.* On a competitor channel, the price they charge for an ad slot is p_k. Advertising a movie on our own channel, meanwhile, comes with the opportunity cost $p_k = V\nu_k$, the revenue forgone by not selling the ad slot at its market price.

On the other hand, the benefit we derive from advertising movie m in ad slot k is the increased ad revenue we can generate by selling ads in movie m for a higher price. This increase in ad revenue we can generate is $V \cdot \Delta\tilde{\nu}_m \cdot b_m$, where $\Delta\tilde{\nu}_m$ is the expected increase in viewership of movie m (as before) and b_m is the number of ad slots in movie m (since we get the benefit of increased viewership across all slots in which the movie shows).

For every ad slot k (on every channel including our own), we have a cost p_k; for every movie m in our schedule, we have $\tilde{r}_m$ and b_m; and for every pair of ad slot and movie we have $\Delta\tilde{\nu}_m \cdot b_m$ expected additional units of ad viewership. Thus, before making any choice of ad slots to use, we can calculate the cost per additional unit of ad viewership (for each *ad slot-movie* pair) as

$$W_{k,m} = \frac{p_k}{\Delta\tilde{\nu}_m \cdot b_m},$$

which is allowed to be infinite if we expect no conversion from this ad slot. Wherever $W_{k,m} \geq V$, the expected payoff of purchasing that ad is negative.

We select ad slots according to the following algorithm:

1. Across all channels, ad slots, and movies, identify $\min\{W_{k,m}\}$ and the corresponding $\hat{k}$, $\hat{m}$ at which this is achieved.
2. If $W_{\hat{k},\hat{m}} \geq V$, STOP (as no profitable ad options remain).
3. Otherwise, mark movie $\hat{m}$ to be advertised in slot $\hat{k}$ and mark this slot as unavailable for other movies.

4. Increase the price of ad slots during movie $\hat{m}$ on our own channel to reflect the increase in viewership (i.e. increase by $V \cdot \Delta\tilde{\nu}_m \cdot b_m$).

5. Update the remaining convertible audience for movie $\hat{m}$: specifically, $r_{\hat{m}} = r_{\hat{m}} - \Delta\tilde{\nu}_m$.

6. Recalculate the expected additional units of viewership from advertising movie $\hat{m}$ for all *other* ad slots k (based on the new value $r_{\hat{m}}$).

7. Recalculate $W_{k,\hat{m}}$ for all ad slots in which we could advertise movie m.

8. Recalculate $W_{k,m}$ for every ad slot *on our own channel* whose price changed in Step 4.

9. Repeat.

Since adverts can only push viewers towards movies playing later in the schedule, we cannot end up with cyclical chains of movies with ad slots advertising each other. If such cycles were possible, it could lead to absurd advertising strategies, for example, in which we use all our ad slots vainly chasing additional viewership and whose value is never realised through the sale of ads. Under our model, this does not happen: the value associated with a viewer of an ad slot k is either realised immediately by the sale of that ad or (the only other option), advertising our own movie generates more value across later ad slots which we do indeed sell.

This greedy strategy is known not to be theoretically optimal (it is easy to construct small counterexamples to illustrate this). It is advantageous, however, as it avoids solving an additional optimisation problem which would be larger and more complex (because of the diminishing returns on adverts for the same movie) than the original scheduling problem. Our results also indicate that it works well in practice.

Ads on our own channel which we do not select to advertise our own movies can then be sold at their market price $V\nu$, accounting for the additional viewership generated by our advertising strategy.

Although we believe this to be the fair market price for these ad slots, should we fail to sell at this price, the maximum benefit we would ourselves derive from using any slot is

$$\max_m \{V \cdot \Delta\tilde{\nu}_m \cdot b_m\},$$

which thus becomes an effective lower bound on the price we should accept to sell this ad slot.

Likewise, if offered more than the maximum benefit we derive from an ad slot, we should agree to the sale, regardless of the selections made by the algorithm above.

2.3.6 How to establish the value V of a unit of ad slot viewership

Recalling our modelling approach for ad slot pricing, establishing the value of one viewer of one ad slot is essential. As the problem is posed in a fictional setting, we attempt to infer this value in a manner consistent with the data provided. This creates interesting dynamics of purchases and sales of ad slots that are reflective of operation of the market, rather than simply a result of arbitrary viewer value we choose to impose.

One major weakness of all our competitors' ad slot pricing strategies is that they do not fully account for viewership. Their prices are calculated in such a way as to ensure a certain level of profitability (assuming all their slots are sold) but they are indifferent to the baseline viewership at different times of the day, except through a 'prime-time factor' which increases ad slot prices between 6pm and 10pm. This means that, relative to their market value, some (indeed many) of our competitor ad slots are mispriced. One initial indicator of this is that the prime-time factor is equal to 1.5, while the viewership at this time is, on average, 1.65 times higher than during the rest of the schedule.

We think it is reasonable, however, that our competitors ad prices *on average* reflect market value. If this were not the case, we argue that their prices would rise or fall until they did. We therefore determine a value for V by summing all our competitor ad slot prices and dividing by the total expected viewership of these ad slots. This results in a value of V of £1.04 million per unit of viewership per ad slot. However, other sources such as [toa24] suggests that this figure is exceedingly high assuming a population of 1 million potential viewers.

It is unfortunately not possible, as the ad-prices of competitors do not reflect viewership, to use this method to infer the values V_g by demographic group. This justifies the simplification we have made of using a single value V per viewer of an ad slot instead. Should better information become available about the values of V_g, these can be immediately incorporated into the model and the prices of ad slots updated as appropriate.

2.3.7 Potential objections to the calculation of the value of a viewer

One might reasonably object to the approach taken to establishing the value of V. In particular, we have criticised competitor ad-pricing strategies (because they do not adequately account for viewership) while simultaneously using those very prices in our own calculation of V. We suggest that, in the absence of external information, our approach is the best way to determine the value of the advertising market in this fictional setting. It means that, while our competitors may misprice particular ad slots, the totals of all their ad revenues would be the same whether priced using our 'market value' or using their own strategy.

3 Results

Model (2.1) generates a one-week movie schedule, which is extended to a 3-month plan using a rolling-forward approach. This approach employs an updated movie database, $\mathcal{M}$, that excludes movies previously shown to avoid repetition. To validate the effectiveness of our model, we first compare the viewership trends of our channel against three competitor channels. Next, we present the derived content strategy in detail. Finally, by integrating the advertising planning strategy outlined in Section 2.3, we propose a comprehensive advertising and pricing strategy.

3.1 Scheduling results

3.1.1 Viewership change

Figure 1 presents the weekly viewership trends for four channels during the first week of operation. Among these, Channel *2* consistently garners the largest audience, whereas Channel *0* attracts the smallest.

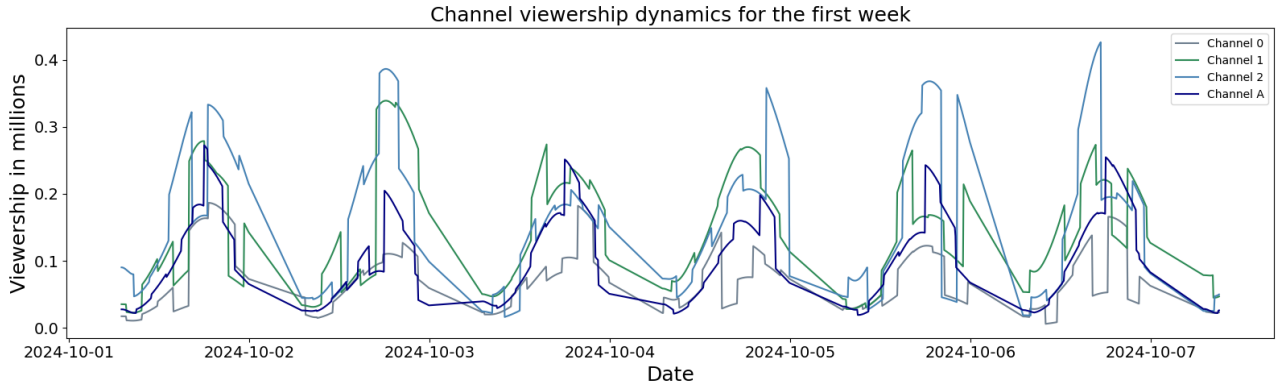


Figure 1: Weekly viewership trends for four channels for week 1

3.1.2 Market share evolution

As mentioned earlier, the television scheduling market is projected to stagnate in the future and therefore, the possibilities of the market growing is slim. Therefore, the only solution to grow is via competing with other channels. Despite the fierce competition in the market, our content scheduling model demonstrated remarkable effectiveness in maintaining our market share. As indicated by Figure 2, we were able to maintain an average of 22% of the market. This highlights our model's ability to create optimal schedules that retain loyal viewers while balancing the complexity of competing content. This is done via a mathematically-driven content strategy discussed further in section 3.2.

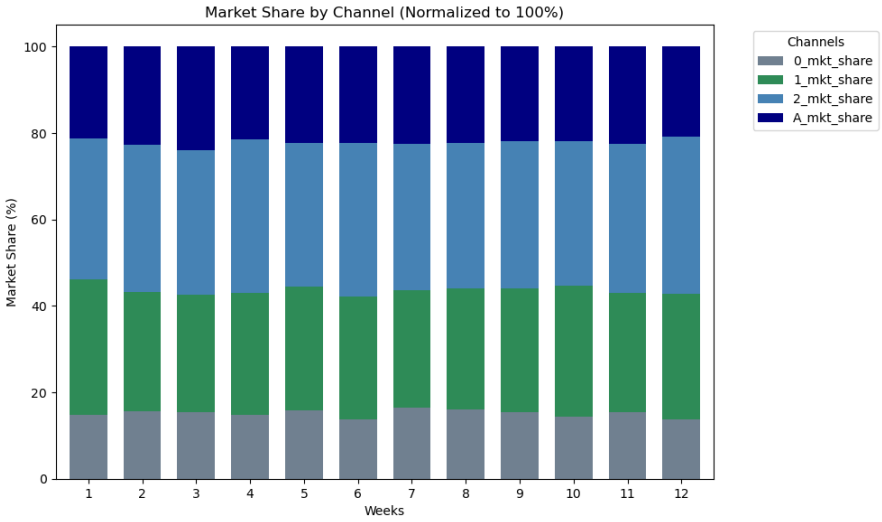


Figure 2: Weekly market share evolution by channel

3.2 Content strategy

3.2.1 Genre and demographic focus

To better understand and structure our results, we performed Principal Component Analysis (PCA) on the movie categories available to license, which revealed a roughly equal distribution across four main thematic buckets represented in Figure 3.

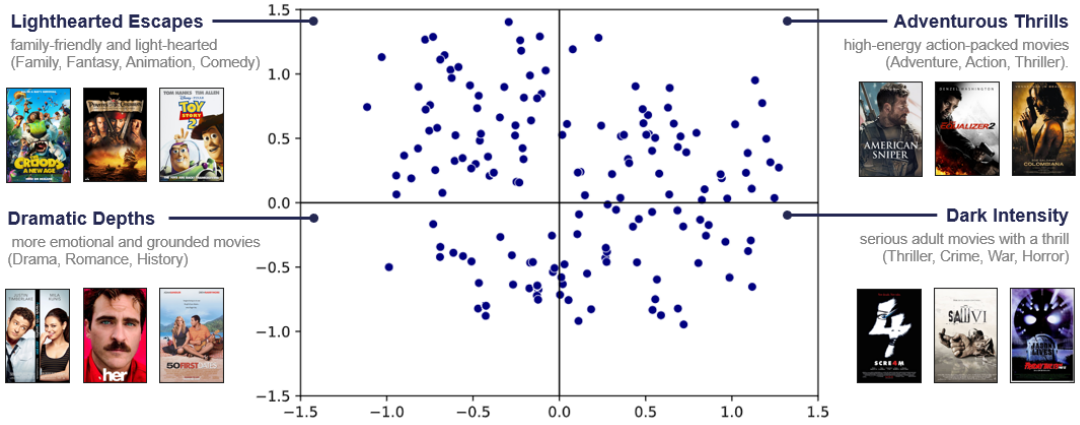


Figure 3: Genre principal component analysis of all movies shown in our schedule

The output of our model also resulted in an equal representation across all categories, recognizing that focusing on a specific genre or category of movies is suboptimal. It important to contextualise this within the rapidly evolving online streaming ecosystem. Lobato's *Rethinking International TV Flows Research in the Age of Netflix* [Lob18], perfectly characterizes the difference in content between the two modes of content distribution. Schedules represent a linear sequences of pre-determined content, while catalogs function as interactive, algorithm-driven databases tailored to individual preferences. This fundamental difference challenges the profitability of niche-focused scheduling in traditional television, as personalized recommendations inherently outperform linear schedules for targeting audiences with specific genre tastes. This is also directly related targeting specific demographics.

When visualising the realised viewership distribution of a representative day [Figure 4], it is evident that adults dominate throughout the entire day, across all channels. Adults are the broadest demographic category of

viewership and therefore, given the discussion above on sub-optimality of niche-focused scheduling in traditional television, it would make sense that the dominance of adult viewership at every single time slot will position them as the primary focus. Our results confirm this hypothesis and indicate that from a demographic perspective, prioritizing adults has the highest impact on viewership. Importantly, our model also incorporates tactical adjustments to accommodate peaks of other demographics at different times of the day, such as on the second week's Thursday 7:00am slot, where we showcase *Toy Story 2*, a movie with a scaled popularity 0.93 for both children and adults.

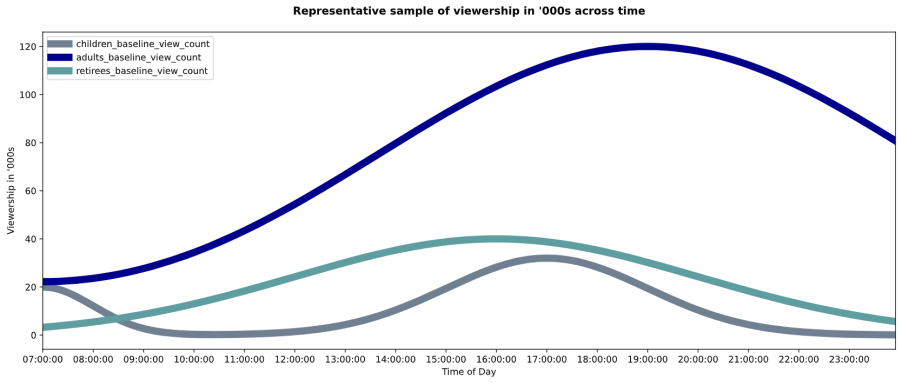


Figure 4: Representative baseline viewership across the day on Channel A

In summary, by maintaining a diversified schedule, we can address the varied expectations of our audience to reinforce our channel's appeal and competitive positioning. This balance is also mirrored by our competitors' channels wherein their schedules also represent a well-rounded selection of categories with a prioritization of the adults demographic, further confirming our analysis.

3.2.2 Release dates and licensing costs

As shown in Table 1, our solution emphasizes the strategic importance of prioritizing newer-than-average productions with higher licensing fees. We have labeled this category as "Premium Fresh Picks" because it captures the intersection of both newer content and higher licensing costs. The higher licensing fee for these productions is directly a function of their strong box office performance and hence, greater appeal and popularity among viewers. Furthermore, when comparing the positioning of our movies to competitors, we observe that they too focus heavily on "Premium Fresh Picks" which validates the results of our model and the importance of prioritizing high-quality, newer content.

	Movie age (years)	Licensing fee (£)
Movie Database	23.2	186,658
Channel A	18.0	373,488
Channel 0	21.7	247,291
Channel 1	21.6	237,696
Channel 2	21.2	240,057

Table 1: Comparison of mean movie age and licensing fee across channels

3.3 Advertising and pricing strategy

The strategy model generates a profit margin of at least 60% in every week of the 12-week period. Profit constitutes net advertising revenues less the costs of licensing fees, so this result already accounts for the model's selection of more popular, more expensive films, which would therefore appear to be a good investment. Full model output appears in Appendix C.

As is apparent from Figure 5, advertising strategy has a significant impact on viewership. The effect is most pronounced for the adult demographic, re-iterating the importance of this group for our strategy overall.

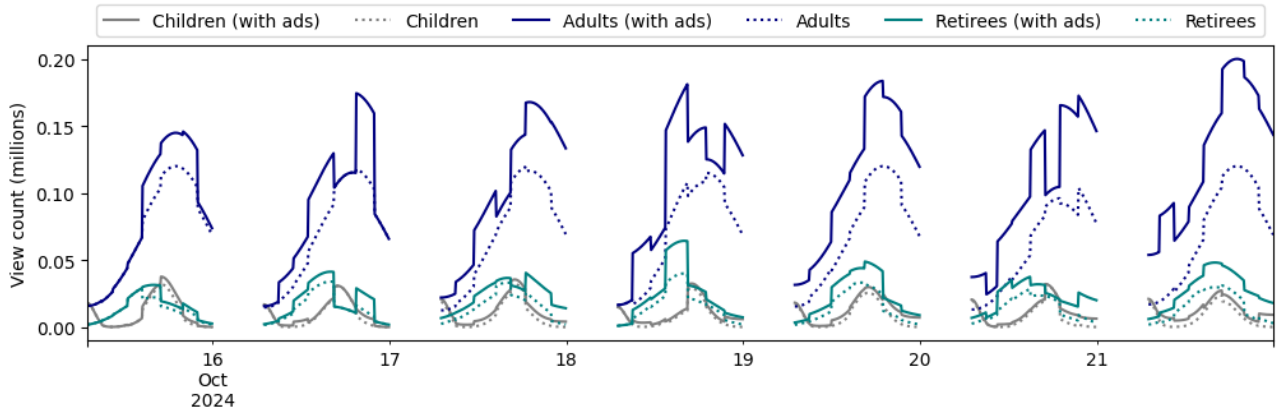


Figure 5: Results of advertising strategy in a typical week

The impact of advertising appears smaller early in the week. This makes sense, since advertising must appear before the target movie, but in the present context, no earlier than the very start of the week.

The advertising and pricing strategy lends itself to an efficient computational implementation. With access to pre-computed conversion rates, it runs in seconds per week (under a minute for the 12-week period).

4 Conclusion

This report tackled the challenge of optimizing the performance of a television channel. We approached the problem by dividing it into two critical components: designing an optimal movie schedule and developing a profit-driven advertising strategy. By splitting the solution into these two parts, we were able to focus on maximizing viewership and profit in distinct yet complementary ways. This decomposition also facilitates an efficient computational implementation. Optimizing the schedule for a single week is typically a matter of a few minutes. With pre-calculated conversion rates, determining the advertising strategy for a single week is a matter of seconds. This implementation, therefore, lends itself to usage in an online (i.e. near-real-time) setting.

In terms of movie selection, we developed a comprehensive mathematical formulation to find the optimal schedule. From that, we derive the following key actionable recommendations for content scheduling:

- Diversify the genre of programs shown throughout the schedule
- Newer, more expensive productions perform better
- Prioritize content for adults and tactically cater to children and retirees

In the advertising strategy, we developed a novel approach to evaluate ad slots based on their conversion rates and expected revenue, allowing us to effectively price each advertisement slot. We were able to successfully leverage the dynamics of the market to our advantage. In summary, we highlight the following key recommendations:

- Capitalize on competitors' mispriced advertisement slots that do not incorporate expected viewership
- Price own advertisement slots based on expected viewership
- Advertise movies within similar genres to maximize conversion into our channel

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Appendices

A Advertisement planning

Building upon the feasible movie scheduling model in Section 2.1, we now extend its functionality to incorporate advertisement planning. This addition enables us to allocate advertisement slots strategically. For simplicity, we assume that every advertisement slot is available for use and that advertising competitor movies on our channel does not negatively impact viewership.

We introduce the following parameters:

- $\mathcal{C}$: the set of channels, where Channels 1, 2, and 3 represent competitor channels, and Channel 4 is our channel.
- $\mathcal{A}$: the set of index of ads slots each day. We assume that there is an ad every 30 mins so 34 ad slots each day.
- $c_{i,j,k,d}$: the cost of advertise movie m at ad slot k on channel j on day d. For the competitor channel, $j \in \{1, 2, 3\}$, the price of advertisement slot is given. For our own channel, $j = 4$, we use the strategy in 2.3.2 to price our own ad slot and we assume that we can sell them with the price.
- b_m: the number of ad slots in movie m.
- C: adverting budget

We add the following decision variables:

- $y_{m,j,k,d} = 0$ if movie m is advertised on channel j at the ad slot k on day d; $= 0$ otherwise $\forall m \in \mathcal{M}, j \in \mathcal{C}, k \in \mathcal{A}$

Thus, with constraints (1b) to (1f), we obtain the full model:

$$\max \quad \sum_{m\in\mathcal{M}} \sum_{t\in\mathcal{T}} \sum_{d\in\mathcal{D}} f_{m,t,d} x_{m,t,d} + \sum_{m\in\mathcal{M}} \sum_{j\in\mathcal{C}} \sum_{d\in\mathcal{D}} \sum_{k\in\mathcal{A}} y_{m,j,k,d} \tag{2a}$$

$$\sum_{m\in\mathcal{M}} y_{m,j,k,d} \leq 1 \qquad \forall j \in \mathcal{C}, k \in \mathcal{A}, d \in \mathcal{D} \tag{2b}$$

$$y_{m,j,k,d} \leq \sum_{t=k}^{T} x_{m,t,d} + \sum_{t\in\mathcal{T}} \sum_{p=d+1}^{D} x_{m,t,p} \qquad \forall m \in \mathcal{M}, j \in \mathcal{C}, k \in \mathcal{T}, d \in \mathcal{D} \tag{2c}$$

$$\sum_{m\in\mathcal{M}} \sum_{j\in\mathcal{C}} \sum_{d\in\mathcal{D}} \sum_{k\in\mathcal{A}} c_{m,j,k,d} y_{m,j,k,d} \leq B \tag{2d}$$

$$\sum_{d\in\mathcal{D}} \sum_{j\in\mathcal{C}} \sum_{k\in\mathcal{A}} y_{m,j,k,d} \leq L \qquad \forall m \in \mathcal{M} \tag{2e}$$

The objective is still to maximize the viewership, but in this model we can attract more viewership by advertising. We assume that our budget on buying ads is limited otherwise we can just buy all the ads slot. This is guaranteed by (2d).

Constraint (2b) ensures that each ad slot can only have one movie. Constraint (2c) ensures that advertise can only before the a movie starts. Constraint (2e) ensures each movie m is not advertised too much.

B Source code

Source code for this project is available at the repository `https://github.com/fahadbahzad/mmcs_project` (private at time of submission).

C Full advertisement strategy results

```
2024-10-01 to 2024-10-08
    Number of adverts: 190
    Net revenue from advertising strategy: 58,206,764
    Total cost of licensing fees: 5,782,757
    Total profit/loss: +52,424,007
    Profit margin: 90%
2024-10-08 to 2024-10-15
    Number of adverts: 220
    Net revenue from advertising strategy: 75,049,644
    Total cost of licensing fees: 14,596,537
    Total profit/loss: +60,453,107
    Profit margin: 81%
2024-10-15 to 2024-10-22
    Number of adverts: 221
    Net revenue from advertising strategy: 70,692,354
    Total cost of licensing fees: 19,548,856
    Total profit/loss: +51,143,498
    Profit margin: 72%
2024-10-22 to 2024-10-29
    Number of adverts: 187
    Net revenue from advertising strategy: 64,043,939
    Total cost of licensing fees: 14,741,283
    Total profit/loss: +49,302,655
    Profit margin: 77%
2024-10-29 to 2024-11-05
    Number of adverts: 203
    Net revenue from advertising strategy: 69,708,686
    Total cost of licensing fees: 18,981,567
    Total profit/loss: +50,727,119
    Profit margin: 73%
2024-11-05 to 2024-11-12
    Number of adverts: 235
    Net revenue from advertising strategy: 72,877,518
    Total cost of licensing fees: 20,542,497
    Total profit/loss: +52,335,021
    Profit margin: 72%
2024-11-12 to 2024-11-19
    Number of adverts: 230
    Net revenue from advertising strategy: 75,716,272
    Total cost of licensing fees: 20,441,273
    Total profit/loss: +55,274,999
    Profit margin: 73%
2024-11-19 to 2024-11-26
    Number of adverts: 213
    Net revenue from advertising strategy: 81,100,735
    Total cost of licensing fees: 28,698,529
    Total profit/loss: +52,402,206
    Profit margin: 65%
2024-11-26 to 2024-12-03
    Number of adverts: 201
    Net revenue from advertising strategy: 72,067,808
    Total cost of licensing fees: 26,060,649
    Total profit/loss: +46,007,159
```

```
    Profit margin: 64%
2024-12-03 to 2024-12-10
    Number of adverts: 201
    Net revenue from advertising strategy: 69,024,916
    Total cost of licensing fees: 27,391,015
    Total profit/loss: +41,633,901
    Profit margin: 60%
2024-12-10 to 2024-12-17
    Number of adverts: 211
    Net revenue from advertising strategy: 72,519,258
    Total cost of licensing fees: 26,754,277
    Total profit/loss: +45,764,982
    Profit margin: 63%
2024-12-17 to 2024-12-24
    Number of adverts: 202
    Net revenue from advertising strategy: 59,830,112
    Total cost of licensing fees: 16,700,565
    Total profit/loss: +43,129,548
    Profit margin: 72%
```